

Lab Report 2: Hypothesis Testing

Women's Representation in the World

DUE: October 15 at 5 PM

You work at the United Nations and have been tasked with improving women's representation in governments around the world. Your boss gives you a dataset including data on 170 countries in the world in 2002. Your boss is particularly interested in understanding regional disparities in women's representation as well as whether changes in electoral systems might improve women's representation. More specifically, she asks you to 1) determine whether there are regional differences in women's legislative representation and 2) whether women's representation is different in proportional representation systems compared to other systems.

You can read in the data "oilwomen.csv" with the following command (or download and import it from CANVAS):

```
dataset <- read.csv("https://raw.githubusercontent.com/ilaydaonder/POLS209/Lab-Report-2/oilwomen.csv")
```

Your boss is interested in the following regions: (a) the Middle East/North Africa (me_nafr) (b) Sub-Saharan Africa (ssafrica) (c) Asia (asia) (d) Latin America (latin). Each of these is coded with a 1 if the country is in that region and a 0 otherwise. For example, Colombia is coded with a 1 for "latin" and a 0 for all other regional variables.

Three other variables are important for your analysis:

- **female_seats**: The percentage of seats in the legislature held by women
- **female_seats_15**: Coded with a 1 if the percentage of seats held by women in the legislature is greater than 15% (the mean)
- **PR**: A binary variable indicating whether the country has a proportional representation system. It is coded 1 if the country has a proportional representation system.

To answer your boss's question, first test the null hypotheses that the percentage of women in the legislature does not differ in each region from the rest of the world (all other countries not in that region). Report your results in the following table. Bold the p-values that are statistically significant.

Region	Region Average	Rest of World Average	Test Statistic	P-Value
Middle East/North Africa				
Sub-Saharan Africa				
Asia				
Latin America				

Then subset the data to create a separate dataset for each region. Test the hypothesis that the average percentage of women in the legislature in a pair of regions is equal. In other words, conduct a difference of means test for each combination of regions. Report your results in the following table. Bold the p-values that are statistically significant.

Region 1	Region 2	Region 1 Mean	Region 2 Mean	P-Value
Middle East/North Africa	Sub-Saharan Africa			
Middle East/North Africa	Asia			
Middle East/North Africa	Latin America			
Sub-Saharan Africa	Asia			
Sub-Saharan Africa	Latin America			
Asia	Latin America			

Finally, test the hypothesis that proportional representation systems and being above average in terms of women's representation (`female_seats_15`) are independent. Report a crosstabulation table and whether you can reject the null based on the test statistic and p-value.

Summarize your results (and include the tables) in a 1-2-page memo. In the memo, interpret the sample means and discuss differences across regions. Do any particular regions tend to be associated with greater women's representation? In what region(s) should the United States focus its efforts at increasing women's representation? Do proportional representation systems seem to improve women's representation? Submit the memo (which should include the tables) and copy/paste your R code at the end.

Grading Rubric (Total: 8 points)

Introduction & Problem Setup (1 point):

- **Full credit (1 point):** The problem is clearly stated, including a full understanding of the variables and regions. The hypotheses are clearly defined.
- **Partial credit (0.5 points):** The problem is somewhat clear but missing essential details (e.g., the definition of one or more variables or hypotheses).
- **No credit (0 points):** The problem setup is incomplete or does not demonstrate an understanding of the variables and hypotheses.

Hypothesis Tests for Regional Differences (2 points):

- **Full credit (2 points):** The hypothesis tests are correctly performed, and results are clearly presented in a table with accurate test statistics and p-values. Significant p-values are bolded.
- **Partial credit (1 point):** Hypothesis tests are mostly correct, but there are minor errors in the calculation or reporting of test statistics and p-values. Some significant p-values may be missing or not bolded.
- **No credit (0 points):** Hypothesis tests are incorrect, or the results are not properly presented. Missing critical information (e.g., p-values, test statistics).

Pairwise Region Comparisons (1.5 points):

- **Full credit (1.5 points):** All pairwise region comparisons are performed correctly with results accurately reported, including p-values. The table is complete, and significant p-values are bolded.
- **Partial credit (0.75 points):** Some pairwise comparisons are missing, or there are minor errors in reporting results. The table is not fully complete or formatted correctly.
- **No credit (0 points):** Pairwise comparisons are incorrect or missing entirely.

Proportional Representation Systems and Women's Representation (1 point):

- **Full credit (1 point):** The correct statistical test (e.g., chi-squared test) is applied to test the hypothesis that PR systems and women's representation are independent. The results are clearly interpreted, and the conclusion about rejecting/accepting the null hypothesis is correct.

- **Partial credit (0.5 points):** The correct test is applied, but there are errors in interpretation, or results are not clearly reported (e.g., no conclusion about rejecting/accepting the null).
- **No credit (0 points):** The test is not performed correctly, or the section is missing critical information.

Memo: Interpretation & Analysis (1 point):

- **Full credit (1 point):** The memo clearly summarizes and interprets the results, including recommendations on which regions to focus on to increase women's representation and whether PR systems improve women's representation. Tables are included and referenced.
- **Partial credit (0.5 points):** The memo is somewhat clear, but missing key details or analysis (e.g., no specific region recommendations or limited discussion of PR systems).
- **No credit (0 points):** The memo is unclear, lacks analysis, or is missing entirely.

R Code Submission & Formatting (0.5 points):

- **Full credit (0.5 points):** The R code is submitted, clearly formatted, and includes comments explaining key steps. The tables are formatted correctly with significant p-values bolded.
- **Partial credit (0.25 points):** The R code is submitted, but is either not well-formatted, lacks comments, or has minor errors in the code structure.
- **No credit (0 points):** R code is missing, not included, or contains major errors that prevent the results from being replicated.

Overall Total: 8 Points